

Bayesian Learning

Lecture 2 - Poisson data. Prior elicitation. Invariant priors.

Mattias Villani

Department of Statistics
Stockholm University



mattiasvillani.com



@matvil



@matvil



mattiasvillani

Lecture overview

- The **Poisson model**
- **Summarizing a posterior distribution**
- **Conjugate priors**
- **Prior elicitation**
- **Jeffreys' prior**

Poisson model

■ Model

$$y_1, \dots, y_n | \theta \stackrel{iid}{\sim} \text{Pois}(\theta)$$

■ Poisson distribution

$$p(y) = \frac{\theta^y e^{-\theta}}{y!}$$

■ Likelihood from iid Poisson sample $y = (y_1, \dots, y_n)$

$$p(y|\theta) = \left[\prod_{i=1}^n p(y_i|\theta) \right] \propto \theta^{(\sum_{i=1}^n y_i)} \exp(-\theta n),$$

■ Prior

$$p(\theta) \propto \theta^{\alpha-1} \exp(-\theta\beta) \propto \text{Gamma}(\alpha, \beta)$$

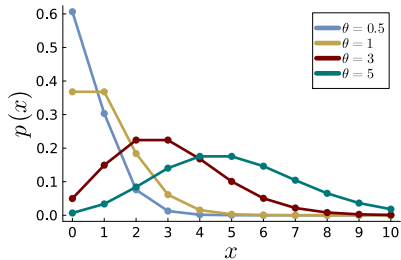
Poisson distribution

$X \sim \text{Pois}(\theta)$
for $X \in 0, 1, 2, \dots$

$$p(x) = \frac{\theta^x e^{-\theta}}{x!}$$

$$\mathbb{E}(X) = \theta$$

$$\mathbb{V}(X) = \theta$$



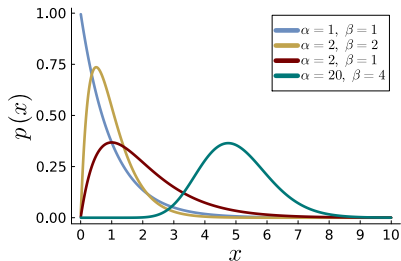
Gamma distribution

$$X \sim \text{Gamma}(\alpha, \beta)$$

$$p(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$$

$$\mathbb{E}(X) = \frac{\alpha}{\beta}$$

$$\mathbb{V}(X) = \frac{\alpha}{\beta^2}$$



Poisson posterior

■ Posterior

$$\begin{aligned} p(\theta|y_1, \dots, y_n) &\propto \left[\prod_{i=1}^n p(y_i|\theta) \right] p(\theta) \\ &\propto \theta^{\sum_{i=1}^n y_i} \exp(-\theta n) \theta^{\alpha-1} \exp(-\theta \beta) \\ &= \theta^{\alpha + \sum_{i=1}^n y_i - 1} \exp[-\theta(\beta + n)], \end{aligned}$$

which is proportional to $\text{Gamma}(\alpha + \sum_{i=1}^n y_i, \beta + n)$.

■ Bayesian updating

Conjugate analysis - Poisson model

Model: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Pois}(\theta)$

Prior: $\theta \sim \text{Gamma}(\alpha, \beta)$

Posterior: $\theta|x_1, \dots, x_n \sim \text{Gamma}(\alpha + \sum_{i=1}^n x_i, \beta + n)$

Example - Number of bids in eBay auctions

■ Data:

- ▶ Number of placed bids in $n = 1000$ eBay coin auctions.
- ▶ Sum of counts: $\sum_{i=1}^n y_i = 3635$.
- ▶ Average number bids per auction: $\bar{y} = 3635/1000 = 3.635$.

■ Prior: $\alpha = 2$, $\beta = 1/2$.

$$\mathbb{E}(\theta) = \frac{\alpha}{\beta} = 4$$

$$\mathbb{S}(\theta) = \sqrt{\frac{\alpha}{\beta^2}} \approx 2.823$$

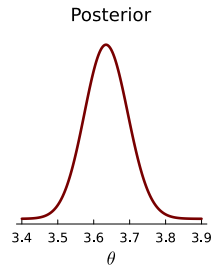
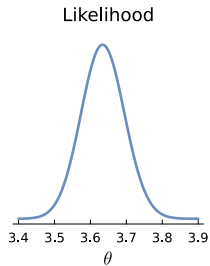
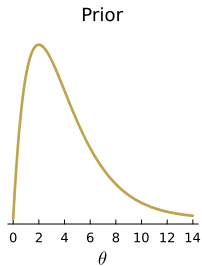
■ Posterior mean and posterior standard deviation

$$\theta|\mathbf{y} \sim \text{Gamma}(\alpha + \sum_{i=1}^n y_i, \beta + n) = \text{Gamma}(3637, 1000.5)$$

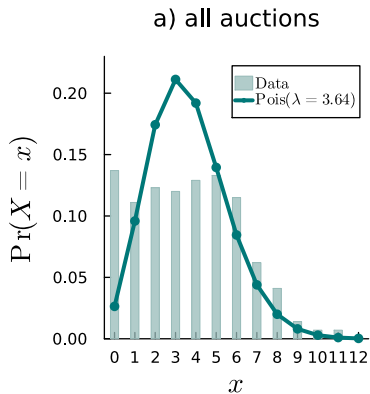
$$\mathbb{E}(\theta|\mathbf{y}) = \frac{\alpha + \sum_{i=1}^n y_i}{\beta + n} = \frac{2 + 3635}{1/2 + 1000} \approx 3.635$$

$$\mathbb{S}(\theta|\mathbf{y}) = \left(\frac{\alpha + \sum_{i=1}^n y_i}{(\beta + n)^2} \right)^{1/2} \approx 0.060$$

eBay data - Posterior of θ



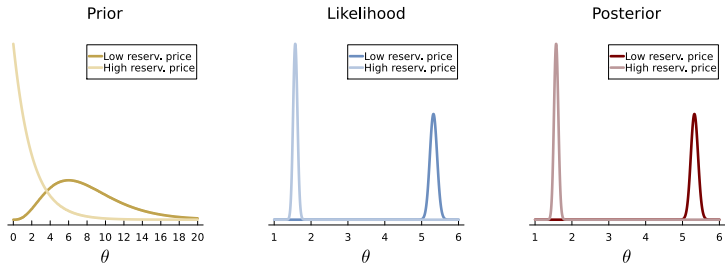
eBay data - model fit at $\theta = \mathbb{E}(\theta|\mathbf{x})$



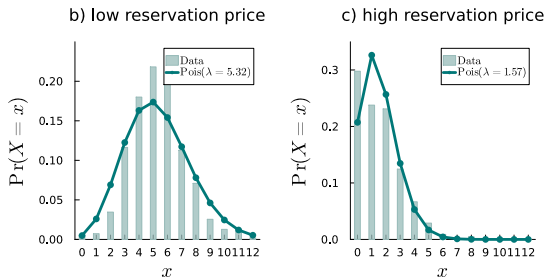
eBay - low/high seller's reservation price

- The data is very heterogenous. Some auctions start with very high reservations prices (lowest price accepted by the seller).
- Split the data into auctions with low/high reservation prices.
- **Low reservation price auctions:**
 - ▶ $n = 550$ eBay coin auctions.
 - ▶ Posterior mean: 5.321 bids.
- **High reservation price auctions:**
 - ▶ $n = 450$ eBay coin auctions.
 - ▶ Posterior mean: 1.576 bids.

eBay data split on reservation price



eBay data - model fit at $\mathbb{E}(\theta|\mathbf{x})$



- Better fits, but still not good enough.
- Even better: **Poisson regression** with reservation price as continuous covariate.

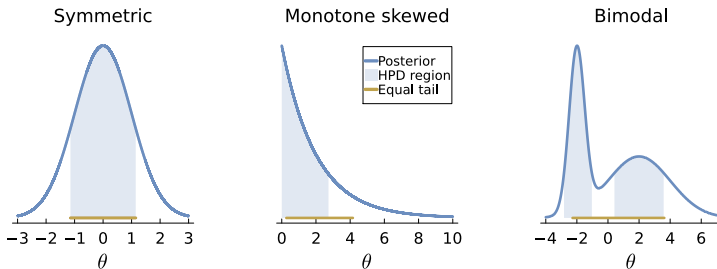
Posterior intervals

- **Bayesian 95% credible interval**: the unknown parameter θ lies in the interval with 0.95 probability **conditional on the observed data**.
- 95% **equal-tail interval**: from 2.5% to 97.5% percentile.
- Approximate 95% **credible interval**

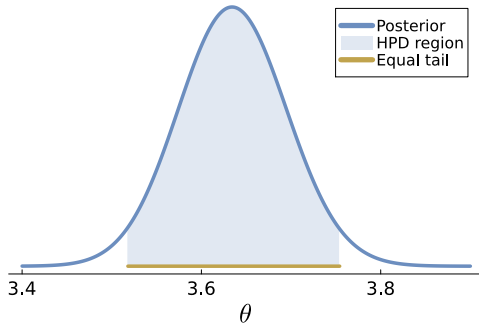
$$\mathbb{E}(\theta|\mathbf{y}) \pm 1.96 \cdot \mathbb{S}(\theta|\mathbf{y})$$

- **Highest Posterior Density (HPD)** interval contains the θ values with highest posterior pdf.

Illustration of different interval types



Credible intervals - eBay auction data



Conjugate priors

- Normal likelihood: Normal prior $\rightarrow$ Normal posterior.
- Bernoulli likelihood: Beta prior $\rightarrow$ Beta posterior.
- Poisson likelihood: Gamma prior $\rightarrow$ Gamma posterior.
- **Conjugate priors**: A prior is conjugate to a model if the prior and posterior belong to the **same distributional family**.

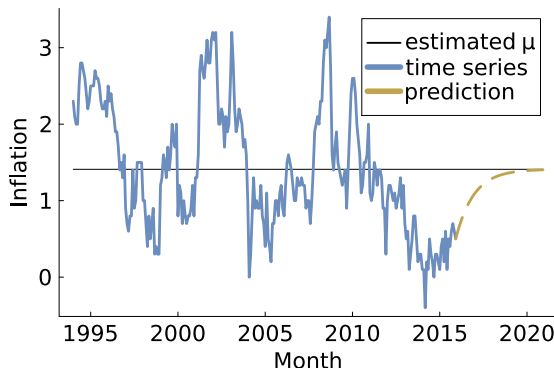
Autoregressive time series model

- **Autoregressive process** of order p - $AR(p)$

$$y_t = \mu + \phi_1(y_{t-1} - \mu) + \dots + \phi_p(y_{t-p} - \mu) + \varepsilon_t, \quad \varepsilon_t \stackrel{iid}{\sim} N(0, \sigma^2)$$

- Unconditional mean: $\mathbb{E}(y_t) = \mu$. Long run forecast attraction.

$$\mathbb{E}(y_{T+h}|y_{1:T}) \rightarrow \mu \text{ as } h \rightarrow \infty.$$



Prior elicitation - AR(p)

■ Autoregressive process

$$y_t = \mu + \phi_1(y_{t-1} - \mu) + \dots + \phi_p(y_{t-p} - \mu) + \varepsilon_t$$

■ **Expert prior** on the unconditional mean: $\mu \sim N(\mu_0, \tau_0^2)$.

■ **Regularization prior** on $\phi_1, \dots, \phi_p$

$$\phi_k \sim N\left(\mu_k, \sigma^2 \frac{\tau^2}{k^2}\right) \text{ independently apriori}$$

- ▶ Prior mean on persistent AR(1): $\mu_1 = 0.8, \mu_2 = \dots = \mu_p = 0$
- ▶ $\mathbb{V}(\phi_k) = \sigma^2 \frac{\tau^2}{k^2}$. Coeff on “longer” lags more likely to be small.

■ Hierarchical prior

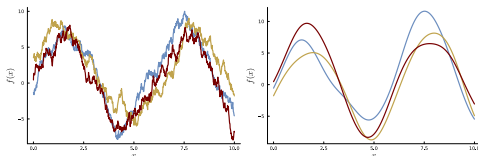
- ▶ Hard to specify τ^2 ? Put a prior on it!
- ▶ $\phi_k | \tau^2 \sim N\left(\mu_k, \sigma^2 \frac{\tau^2}{k^2}\right)$ and $\tau^2 \sim \chi_\nu^2$.
- ▶ Gives a posterior on global shrinkage τ^2 .

Prior elicitation

Smoothness priors

- ▶ a version of regularization priors
- ▶ nonlinear regression function $f(\cdot)$ is believed to be smooth

$$y = f(x) + \varepsilon$$



Noninformative priors

- ▶ **Uniform:** $\theta \sim \text{Beta}(1, 1)$.
Issue 1: same as prior sample with one success and one failure.
Issue 2: not uniform for $\phi = \log \frac{\theta}{1-\theta}$.
- ▶ **Zero prior sample size:** $\theta \sim \text{Beta}(\epsilon, \epsilon)$ with $\epsilon \downarrow 0$.
Posterior $\rightarrow \text{Beta}(s, f)$.
Issue: posterior is improper if $s = 0$ or $f = 0$.

Invariant prior

■ Observed information

$$J_{\mathbf{x}}(\hat{\theta}) = - \frac{d^2 \ln p(\mathbf{x}|\theta)}{d\theta^2} \Big|_{\theta=\hat{\theta}}$$

■ Fisher information

$$I(\theta) = E_{\mathbf{x}|\theta} (J_{\mathbf{x}}(\theta))$$

■ Jeffreys' rule to construct prior

$$p(\theta) = I(\theta)^{1/2}.$$

■ Invariance under 1:1 parameter transformation $\phi = g(\theta)$.

Example: $\phi = \log \frac{\theta}{1-\theta}$.

▶ Specify $p_{\theta}(\theta)$ directly

▶ Specify $p_{\phi}(\phi)$ and then obtain $p_{\theta}(\theta) = p_{\phi}(g^{-1}(\theta)) \left| \frac{dg^{-1}(\theta)}{d\theta} \right|$.

Invariant prior

■ Observed information

$$J_{\mathbf{x}}(\hat{\theta}) = - \frac{d^2 \ln p(\mathbf{x}|\theta)}{d\theta^2} \Big|_{\theta=\hat{\theta}}$$

■ Fisher information

$$I(\theta) = E_{\mathbf{x}|\theta} (J_{\mathbf{x}}(\theta))$$

■ Jeffreys' rule to construct prior

$$p(\theta) = I(\theta)^{1/2}.$$

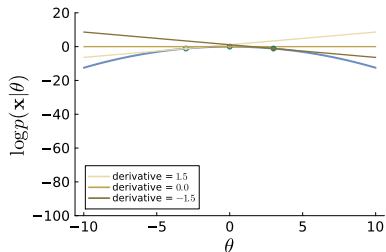
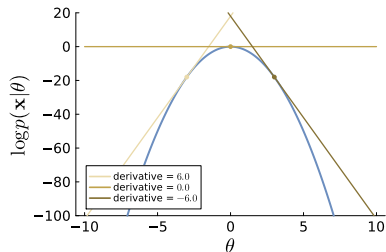
■ Invariance under 1:1 parameter transformation $\phi = g(\theta)$.

Example: $\phi = \log \frac{\theta}{1-\theta}$.

▶ Specify $p_{\theta}(\theta)$ directly

▶ Specify $p_{\phi}(\phi)$ and then obtain $p_{\theta}(\theta) = p_{\phi}(g^{-1}(\theta)) \left| \frac{dg^{-1}(\theta)}{d\theta} \right|$.

Second derivative measures curvature



Jeffreys' prior for negative binomial sampling

- Jeffreys' prior:

$$n|\theta \stackrel{iid}{\sim} \text{NegBin}(s, \theta).$$

$$\ln p(\mathbf{x}|\theta) = \ln \binom{n-1}{s-1} + s \ln \theta + f \ln(1 - \theta)$$

$$\frac{d^2 \ln p(\mathbf{x}|\theta)}{d\theta^2} = -\frac{s}{\theta^2} - \frac{f}{(1 - \theta)^2}$$

$$I(\theta) = \frac{s}{\theta^2} + \frac{E_{n|\theta}(n - s)}{(1 - \theta)^2} = \frac{s}{\theta^2} + \frac{s/\theta - s}{(1 - \theta)^2} = \frac{s}{\theta^2(1 - \theta)}$$

- Thus, the Jeffreys' prior is

$$p(\theta) = |I(\theta)|^{1/2} \propto \theta^{-1}(1 - \theta)^{-1/2} \propto \text{Beta}(\theta|0, 1/2).$$

- Jeffreys' prior is **improper**, but the posterior is proper:
 $\theta|n \sim \text{Beta}(s, f + 1/2)$ which is proper since $s \geq 1$.
- Jeffreys' prior **violates the likelihood principle** because $I(\theta)$ is sampling-based.